

WORKSHEET - NAPLAN Year 9

Number and Algebra Basic Calculations and Algebra (9)

Teacher: Joanne Rust

Exam Equivalent Time: 8.75 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. Number and Algebra, NAP-55610

Byron earns \$16 per hour.

How much will he be paid for working 8 hours?

\$2 \$32 \$128 \$216

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2. Number and Algebra, NAP-90964

If $p = 7$, what is the value of $2p$?

14 27 49 98

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3. Number and Algebra, NAP-67442

At 7 am the temperature in Merewether was 15.9°C .

At midday it was 12.8°C warmer.

A 7 pm it was 13.9°C cooler than at midday.

What was the temperature at 6 pm?

14.8°C 15°C 17°C 42.6°C

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4. Number and Algebra, NAP-26176

What is the value of the expression $3 + (-13) - (-6)$?

-16 -4 18 26

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5. Number and Algebra, NAP-79193

Which expression is equivalent to $6x + 12$?

$3(2x + 4)$ $3(2x + 9)$ $6(x + 6)$ $6(x + 12)$

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6. Number and Algebra, NAP-20296

Which of the following has the same value as 7^{-11} ?

-7^{-11} $7 \times (-11)$ $\frac{1}{-7^{11}}$ $\frac{1}{7^{11}}$

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7. Number and Algebra, NAP-02719

A red blood cell and a white blood cell have a combined total mass of 1.3×10^{-10} grams.

If the red blood cell has a mass of 3×10^{-11} grams, what is the mass of the white blood cell?

☐ 1×10^{-10} grams

☐ 1×10^{-11} grams

☐ 1.7×10^{-10} grams

☐ 1.7×10^{-11} grams

Worked Solutions

1. Number and Algebra, NAP-55610

$$\begin{aligned}\text{Pay} &= 8 \times 16 \\ &= \$128\end{aligned}$$

2. Number and Algebra, NAP-90964

$$\begin{aligned}p &= 7 \\ 2p &= 2 \times 7 \\ &= 14\end{aligned}$$

3. Number and Algebra, NAP-67442

$$\begin{aligned}\text{Temperature at 6 pm} \\ &= 15.9 + 12.8 - 13.9 \\ &= 14.8^\circ\text{C}\end{aligned}$$

4. Number and Algebra, NAP-26176

$$\begin{aligned}3 + (-13) - (-6) &= -10 + 6 \\ &= -4\end{aligned}$$

5. Number and Algebra, NAP-79193

$$3(2x + 4) = 6x + 12$$

6. Number and Algebra, NAP-20296

$$\frac{1}{7^{11}}$$

7. Number and Algebra, NAP-02719

$$\begin{aligned}1.3 \times 10^{-10} - 3 \times 10^{-11} \\ &= 1.0 \times 10^{-10} \text{ (by calculator)}\end{aligned}$$

Worked Solutions